

1. To retrieve / view the data from back end

```
import java.io.*;
import java.sql.*;
class MysqlCon
{
    public static void main(String args[])
    {
        try
        {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con=DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/mag?useSSL=false&serverTimezone=UTC","root","root");

                //here db is database name, root is username and password

            Statement stmt=con.createStatement();
            ResultSet rs=stmt.executeQuery("select * from stud");
            while(rs.next())
            System.out.println(rs.getInt(1));
            con.close();
        }
        catch(Exception e){ System.out.println(e);}
    }
}
```

Question 1 – FileInputStream & FileOutputStream

```
C:\Java>java SingleDataStream
Data read from file: A
```

```
C:\Java>java BulkDataStream
Data read from file: Welcome to Java File Handling
```

2. To update / retrieve the data from backend

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.Statement;
import java.io.*;
import java.sql.*;

class SB
{
    public static void main(String args[]) throws IOException
    {
        try
        {
            Class.forName("com.mysql.jdbc.Driver");

            Connection con=DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/mag?useSSL=false","root","root");

            Statement stmt=con.createStatement();

            String sql = "create table bank(accno int not null,
name varchar(20), balance double(10,2), primary key (accno))";
```

```
stmt.executeUpdate(sql);

String query = "INSERT INTO bank (accno, name,
balance) VALUES (?, ?, ?)";

PreparedStatement ps =
con.prepareStatement(query);

    ps.setInt(1, 101);
    ps.setString(2, "chandru");
    ps.setDouble(3, 10000.00);
    ps.addBatch();

    ps.setInt(1, 102);
    ps.setString(2, "mouli");
    ps.setDouble(3, 6000.00);
    ps.addBatch();

    ps.setInt(1, 103);
    ps.setString(2, "eswar");
    ps.setDouble(3, 7000.00);
    ps.addBatch();

ps.executeBatch();

ResultSet rs = stmt.executeQuery("select * from
bank");

System.out.printf("accno" + "\t" + "name" + "\t\t" +
"balance" );
```

```

        System.out.println();
        while(rs.next())
        {
            System.out.printf(rs.getInt(1)+ "\t");
            System.out.printf(rs.getString(2) + "\t\t");
            System.out.println(rs.getDouble(3));
        }

        // deposit operations for accno 101
        sql = "update bank set balance = balance + 5000
where accno =101";
        stmt.executeUpdate(sql);

        // withdraw operations for accno 102
        sql = "update bank set balance = balance - 3000
where accno =102";
        stmt.executeUpdate(sql);

        sql = "delete from bank where accno = 103";
        stmt.executeUpdate(sql);

        rs = stmt.executeQuery("select * from bank");
        System.out.println();
        System.out.printf("accno" + "\t" + "name" + "\t\t" +
"balance" );

        System.out.println();

```

```

        while(rs.next())
        {
            System.out.printf(rs.getInt(1)+ "\t");
            System.out.printf(rs.getString(2) + "\t\t");
            System.out.println(rs.getDouble(3));
        }
        con.close();
    }
    catch(Exception e){ System.out.println(e);}
}
}

```

Question 2 – FileReader & FileWriter

```
C:\Java>java SingleDataReaderWriter
```

```
Character read from file: A
```

```
C:\Java>java BulkDataReaderWriter
```

```
Data read from file:
```

```
Java File Reader and Writer Example
```

Using AWT, providing user interface and event handling initiates user interaction with back end

```
import javax.swing.*;
```

```
import java.sql.*;
```

```
import java.awt.event.*;
```

```
import java.awt.*;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.ResultSetMetaData;
import java.sql.Statement;
import java.io.*;
import java.util.*;

class myframe1 extends JFrame implements ActionListener
{
    JLabel l1,l2,l3,l4,l5,l6,l7;
    JComboBox jc1,jc2;
    JTextField t1,t2,t3,t4;
    JButton b1;

    public myframe1()
    {
        l1=new JLabel("Name :");
        l1.setBounds(100,100,150,50);
        this.getContentPane().setLayout(null);
        this.getContentPane().add(l1);

        l2=new JLabel("Father's Name:");
        l2.setBounds(100,150,150,50);
        this.getContentPane().add(l2);
```

```
l3=new JLabel("Date of Birth:");  
l3.setBounds(100,200,150,50);  
this.getContentPane().add(l3);
```

```
l4=new JLabel("Sex:");  
l4.setBounds(100,250,150,50);  
this.getContentPane().add(l4);
```

```
l5=new JLabel("Address:");  
l5.setBounds(100,300,150,50);  
this.getContentPane().add(l5);
```

```
l6=new JLabel("Qualification:");  
l6.setBounds(100,350,150,50);  
this.getContentPane().add(l6);
```

```
t1=new JTextField();  
t1.setBounds(250,110,150,25);  
this.getContentPane().add(t1);
```

```
t2=new JTextField();  
t2.setBounds(250,160,150,25);  
this.getContentPane().add(t2);
```

```
t3=new JTextField();  
t3.setBounds(250,210,150,25);  
this.getContentPane().add(t3);
```

```
jc1=new JComboBox();  
jc1.addItem("M");  
jc1.addItem("F");  
jc1.setBounds(250,260,150,25);  
this.getContentPane().add(jc1);
```

```
t4=new JTextField();  
t4.setBounds(250,310,150,25);  
this.getContentPane().add(t4);
```

```
jc2=new JComboBox();  
jc2.addItem("M.E.,");  
jc2.addItem("Ph.D.,");  
jc2.setBounds(250,360,150,25);  
this.getContentPane().add(jc2);
```

```
b1=new JButton("Submit");  
b1.setBounds(350,400,150,25);  
this.getContentPane().add(b1);
```



```
l7=new JLabel("");  
l7.setBounds(400,500,150,200);  
this.getContentPane().add(l7);
```

```
b1.addActionListener(this);  
}
```

@Override

```
public void actionPerformed(ActionEvent ae)  
{  
    String x,y;  
    try  
    {
```

```
        Class.forName("com.mysql.cj.jdbc.Driver");
```

```
        Connection
```

```
conn=DriverManager.getConnection("jdbc:mysql://localhost:3306/m  
ag?useSSL=false&serverTimezone=UTC","root","root");
```

```
        Statement stmt=conn.createStatement();
```

```
        x = String.valueOf(jc1.getSelectedItem());
```

```
        y= String.valueOf(jc2.getSelectedItem());
```

```
        String s = "insert into Biodata  
values('"+t1.getText()+"','"+t2.getText()+"','"+t3.getText()+"','"+x+"','"+t4  
.getText()+"','"+y+"')";
```

```
        stmt.executeUpdate(s);
```

```

        stmt.close();
        conn.close();
        l7.setText("Record Inserted");
    }
    catch(ClassNotFoundException ce)
    {
    }
    catch(SQLException e)
    {
    }

}
}

class myapp1{
    public static void main(String arg[]) throws IOException,
    SQLException {
        myframe1 f=new myframe1();
        f.setTitle("Welecome to enter data for Biodata");
        f.show();
    }
}

```

Question 3 – Number of Lines & Characters

```
C:\Java>java FileCount  
Number of characters: 58  
Number of lines: 3
```

Question 4 – SequenceInputStream

```
C:\Java>java SequenceFileStream  
Files successfully combined.
```

```
output.txt:
```

```
This is File 1.
```

```
This is File 2.
```